

Algebra 1
Unit 7
Lesson 7 Practice

Name Answer key
Date _____
Period _____

For questions 1-2, consider the perfect square trinomial $25x^2 + 30x + 9$.

1. Rewrite the trinomial in the form $a^2 + 2ab + b^2$.

$$(5x)^2 + 2(5x)(3) + (3)^2$$

2. Factor the perfect square trinomial.

$$(5x + 3)^2$$

For questions 3-4, consider the perfect square trinomial $9x^2 - 42x + 49$.

3. Rewrite the trinomial in the form $a^2 - 2ab + b^2$.

$$(3x)^2 - 2(3x)(7) + (7)^2$$

4. Factor the perfect square trinomial.

$$(3x - 7)^2$$

For questions 5-8, determine whether the quadratic expression is a perfect square trinomial. Then, factor the expression.

5. $w^2 - 24w + 144$

Yes.
 $(w - 12)^2$

6. $p^2 + 2pq + q^2$

Yes.
 $(p + q)^2$

7. $49v^2 - 25$

No.
 $(7v - 5)(7v + 5)$

8. $16y^2 - 256y + 32$

No.
 $16(y^2 - 16y + 2)$

For questions 9-10, factor the trinomial.

9. $36a^2 - 60a + 25$

$(6a - 5)^2$

10. $p^2 - 18p + 81$

$(p - 9)^2$